

Classwork 07//KL Divergence in DDPM

Student Name: _____ **Points:** 30

A Number: _____ **Due Date:** March 23, 2026

KL Divergence for Denoising Diffusion Model (DDPM) is given by

$$\text{KL}(q, p) = \underbrace{\mathbb{E}_{q(\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_T)} \left[-\log p(\mathbf{x}_T) - \sum_{t=1}^T \log \frac{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)}{q(\mathbf{x}_t | \mathbf{x}_{t-1})} \right]}_{=L} + \text{constant}$$

Show that L can be written as

$$\mathbb{E}_{q(\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_T)} \left[\text{KL}(q(\mathbf{x}_T | \mathbf{x}_0) || p(\mathbf{x}_T)) + \sum_{t>1} \text{KL}(q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) || p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)) - \log p_\theta(\mathbf{x}_0 | \mathbf{x}_1) \right]$$

Use the fundamental definition of KL divergence for proving.